

EDITORIAL

Ten tips for writing a brilliant PhD thesis – and enjoying the process

The prospect of writing a PhD thesis – which may exceed 100,000 words – will inevitably seem like a daunting task. After all, this is a document that must encapsulate all of the hard work that has gone into generating and analysing a mass of information collected over a two-to-three-year period, will be scrutinised by examiners who are authorities in the field, and will be incorporated in the hallowed archives of a university library.

I clearly recall the process of writing my own PhD thesis as the most enjoyable aspect of my candidature. I basically saw my thesis writing as a fascinating journey of discovery. This is because the process of writing, constructing arguments, and placing my work in the context of the existing literature encouraged me to think critically about what I had done, and allowed me to appreciate the true impact and implications of my work.

Although I am referring here to writing a PhD thesis, the principles I espouse also apply to writing an MSc thesis, or indeed any other form of lengthy scientific dissertation. My tips primarily refer to writing a thesis in a medical/scientific field, although much of what I say would also apply to other genres.

As for my credentials in respect of writing this editorial: I have authored a PhD and a DSc thesis, supervised 17 MSc and 28 PhD candidates to successful completion, and examined 6 MSc, 21 PhD and 3 DSc theses. I have also delivered a number of university seminars on thesis writing.

So, to assist those commencing PhD candidature – and especially those about to write up their PhD thesis – I offer the following 10 tips on how to best go about this task. Your aim should be to generate a brilliantly written thesis (and hopefully one that is also scientifically brilliant), and more importantly, enjoy the process.

Tip #1: Arrange a comfortable, quiet writing space

Writing a PhD thesis required a good deal of planning, thinking and quiet reflection. This is a process that could take between six and 12 months of essentially full-time writing, assuming that you are adopting the approach of substantially writing your thesis as the last phase of your candidature, after experimental work has largely concluded. An alternative and equally valid approach is to write sections throughout your candidature. Although the comments below primarily apply to my preferred approach (block writing at the end), they are generally relevant to both.

There will be phases of rapid writing when ideas are flowing, and slow, interrupted writing when trying to formulate detailed and complex arguments. To write effectively, a key requirement is to locate yourself in a light and airy, ventilated, uncluttered workspace (an uncluttered workspace means an uncluttered mind). This space will almost certainly not be your laboratory or communal workspace shared with other graduate students at your university.

The best arrangement is to set up a comfortable office or workstation at your place of residence, preferably away from

others in the household. Invest in a very comfortable office chair because you will be spending virtually all your writing time sitting on it. Have a good supply of coffee or other favourite drinks at hand, as well as a small selection of snacks.

So basically, I am advising that you write your thesis largely at home. Be sure, however, to go into your workplace for at least a few hours per week to stay connected with fellow students, academic staff and most importantly, your supervisor/s.

Tip #2: Study the university thesis submission requirements

Familiarise yourself with the rules, regulations and guidance for PhD candidature at your university, especially, in this context, those relating to the actual thesis. First and foremost, be clear on timelines and deadlines for submission. Be aware of formatting requirements for the thesis, and whether it is to be submitted as a traditional bound paper thesis, electronic file, or both. Note any advice on the size and type of font, headers, footers and margin sizes. Be aware of limits in terms of the number of words, pages or references, as they may apply to specific sections (e.g. abstract) or the entire thesis. Check if the reference list must be prepared in an advised or mandated style, and be sure to adopt that (see Tip #8).

Tip #3: Construct a detailed thesis table of contents

A useful approach to thesis writing that I discovered myself, and have always advised my higher degree students, is to start by writing a very detailed table of contents. This task alone may take a few days to complete, and run into many pages. The table of contents should list all the planned chapters, and within each chapter, devise a series of sections, headings and sub-headings, extending down to possibly three or four levels of hierarchy. For ease of navigation during the writing process, hyperlink the table of contents to the body of the thesis.

The reason for constructing a detailed table of contents at the beginning is that it transforms the seemingly arduous task of writing a 100,000-word thesis into an incremental process of writing a series of 200-word sections. Let me explain. If your table of contents was comprised of, say, 50 sections/subsections, then each section, on average, will be about 200 words. In isolation, writing a brief 200-word section ought to be a very achievable and enjoyable task. If you set a target of writing, say, 5 sections per day (1,000 words), your thesis could be completed in 100 writing days.

As you proceed with your thesis writing, it may well become apparent that the structure of the table of contents needs to be modified, and that's fine. The point is that devising a table of contents up front gives structure and purpose to your thinking and writing, turning a seemingly arduous task into an enjoyable process of authoring a series of well-defined and contextualised 200-word sections.

Another benefit of devising a detailed table of contents at the start is that it helps structure your thinking as you progress through the writing process, and can serve as a repository for any ideas that come to mind, which can simply be inserted as a brief reminder note into the relevant section, for subsequent consideration.

Tip #4: Bash out that first sentence

'Writer's block' is a well-worn cliché, but it is a common problem that you may well encounter at the start of your thesis writing, as well as at various junctures thereafter. Without doubt, the hardest part of producing a thesis is writing the first sentence of any given section or chapter. So, if you find yourself in a deep, depressive cycle of indecision and hesitancy when trying to write a new section, you need to find a circuit-breaker. My advice is simple: sit back, take a deep breath, then just 'bash it out'. You will often find that writing that first sentence will trigger a subsequent train of thought, and the next sentence will flow on much more easily. You can then capitalise on this momentum and continue writing until the ideas dry up.

Almost certainly, what you have bashed out will be imperfect, with spelling and grammatical errors, but that doesn't matter. It means you have made a start and your thought processing has been kick-started. The initial effort can be subsequently refined. Adopting this 'just bash it out' strategy will help you side-step writer's block and maintain good momentum.

Tip #5: Make the introduction tantalising and the conclusion decisive

The introduction to your thesis (as distinct from the abstract, discussed in Tip #10) should be designed to capture the imagination of the reader, and provide a series of tantalising hints as to what is to come. It should put your intended work into a broad perspective, unlike the literature review (see Tip #6), which places your work into a more detailed context in relation to past and current thinking.

At the same time, the introduction must contain formal structural elements. The aims of your thesis should be clear and simply stated, perhaps as a bulleted list of succinct points. Following on from this should be a clear set of objectives.

The aims should be followed by your hypothesis, and these two elements should not be confused. For example, an *aim* might be to 'examine the height of males and females', to test a *hypothesis* that 'males are taller than females'. The introduction should explain why your work is important and how it will advance knowledge in the field.

Every chapter should finish with a brief 'Conclusions' section, and the entire thesis should finish with a 'Conclusions' chapter, with the general approach to either being the same; that is, conclusions should be informative, decisive, and explain the meaning of your work. Draw together your key points and how they contribute to knowledge in the field, reflecting back on your aims and hypotheses where relevant.

After outlining your key conclusions, you have licence to stray beyond the proven findings of your thesis and speculate on future research directions. Try and explain what gaps your own research has exposed, and what needs to be done next to fill these gaps.

Another important component of a concluding chapter is the analysis of the strengths and weaknesses in your work. You will certainly get credit for demonstrating a capacity to engage in self-reflection and critically analyse your own work.

Tip #6: Keep the literature review focussed

Strive to keep your literature review focussed to your specific field. For a PhD thesis, it is acceptable to assume a certain amount of prior knowledge. So, for example, if you are writing about the causes of dry eye disease, there is no need to provide a detailed account of the anatomy of the entire eye; instead, focus on just the structures of the anterior eye relating directly to dry eye disease.

Needless to say, an essential prerequisite to writing a sensible, informed and relevant literature review is to know the literature surrounding your field of interest. In this regard, there is no substitute for reading widely, which is a process that will begin when defining the subject matter or your thesis, and will continue throughout your candidature. The literature review must not be an uncritical – and likely boring – regurgitation of previously published work; it must be a critical account, highlighting important findings and perhaps challenging others, and adjudicating any past controversies or disagreements. Identify gaps in knowledge, inasmuch as they have directed your intended research. A useful way to distil down a complex body of the literature is to summarise it in tabular form.

Familiarise yourself with the search protocols required for your bibliographic data base of choice, conduct targeted searches, and compile relevant references in a bibliographic referencing database that is capable of storing abstracts and, if possible, PDF files of complete papers.

Tip #7: Develop a disciplined scientific writing style

The 'scientific style' is a very specific and somewhat formal style of writing that must be adopted for a PhD thesis in the sciences. To some extent, new PhD students would have been exposed to this writing style during their undergraduate studies, and it will become more familiar in the course of reading the literature throughout their candidature. Here are a few guidelines that I believe are important to adhere to, couched in terms of what to avoid when writing your thesis:

- Avoid jargon or exotic words.
- Avoid superlatives (e.g. 'fantastic', 'phenomenal', 'tremendous', 'horrendous').
- Avoid using the passive voice.
- Avoid the use of first person (i.e. 'I', 'we', 'our', 'us').
- Avoid long, heavily punctuated, convoluted sentences; instead, strive for short, sharp sentences.
- Avoid the apostrophe form (e.g. 'vision of the subject' rather than 'the subject's vision').
- Avoid errors in verb/noun agreement (e.g. 'data were analysed', not 'data was analysed').

Acronyms/abbreviations should be used infrequently, and only for long terms or phrases that are used frequently. Write out an acronym/abbreviation in full the first time it is used. If you end up using many acronyms/abbreviations, provide a list of these in alphabetical order, alongside the full terms, in the introductory material.

Adopt a logical grammatical structure: one point per sentence; one idea per paragraph; one theme per section;

and one concept per chapter. Try to maintain a logical flow of ideas. Liberally use 'connectors' throughout your thesis, such as 'as discussed in Chapter 6' or '(see Chapter 2 for details)'. As well as assisting the reader to navigate your thesis, this strategy will give your thesis an integrated feel.

Be sure to leave time for re-reading your work, for two reasons. First, a second reading will help you identify any argument gaps, inconsistencies, redundancies or overlap, that of course will need to be corrected. It may be helpful to generate a check-list of terms or style conventions that need to be consistent throughout the thesis. Check for inconsistencies in headings and subheadings.

Second, a re-reading will provide you with the opportunity to correct any spelling/grammatical errors that the automated checkers may have missed; these systems are not perfect. You are not perfect either, so if possible have a friend or colleague read your thesis; they do not have to be an expert in the field to check for spelling/grammatical errors and appropriate sentence/paragraph structure.

Perhaps most importantly, don't trust 'The Cloud', and constantly back up your work.

Tip #8: Be fastidious in your referencing

Use the referencing style recommended/mandated by your university. Whatever style you do use, make sure a consistent approach is adopted throughout your thesis. It is important to be pedantic, paying attention to such issues as whether the journal page range is presented in full or abbreviated form, whether a full stop, or comas or no punctuation is used after the initials of an author name, etc. A perfectly formatted bibliography will be viewed as sharp, disciplined scientific presentation; conversely, sloppy referencing will be viewed as sloppy science, which is not the impression you would wish to convey to an examiner.

Use an automated reference manager, but be wary of the formatting that is finally produced. These reference managers are not perfect, and will sometimes, for example, set the title of a paper as 'First Letter Of All Words Capitalised' instead of 'Sentence case'. The reason for imperfect automated referencing may be flaws in the algorithm used; inconsistencies in the way references are entered by the publisher into the database; flaws in the way a given database stores references; or your own error when manually inserting a reference in your reference manager. It is therefore essential to thoroughly check your bibliography *after* it has been automatically formatted.

A common error when preparing a bibliography is to refer to a paper in the text of your thesis which does not appear in your reference list, or conversely, listing a reference in your bibliography that is not mentioned anywhere in your thesis. These issues can generally be avoided by using a reference manager.

Err on the side of referencing the most recent literature, although older pivotal papers or classic papers should be referenced if relevant.

Tip #9: Craft a succinct and informative title

Although you will adopt a working title for your thesis at the start of your candidature, it necessarily will have been formulated before the results are known and the full impact of your research is realised. Research can take twists and

turns, and it is unlikely your research path will end up where you had initially intended. Therefore, it is a good idea to sit down, and with the benefit of hindsight, reconsider the title of your thesis to ensure it accurately reflects its content.

As to the length of the title – there is a natural tension between being succinct versus informative. The title should be interesting and serve as an invitation to the reader to see what is inside. Make sure to include key words or phrases in the title. Do not use abbreviations or jargon. The title should be as brief as possible, and certainly no longer than about 12 words.

Tip #10: Ensure your abstract captures the 'big picture'

I have deliberately left the most important tip until last – the reason being that writing the abstract should be your very last piece of thesis writing. The abstract serves as an executive summary; it needs to capture the 'big picture', but also must briefly explain 'how, why, what, and so what', generally in only 300 to 500 words (or whatever word count advised by your university guidelines).

Your abstract is critically important for a number of reasons. Some universities send the abstract of a thesis to prospective examiners, to gauge their interest in assessing the work. When it comes to actually examining a thesis, the abstract is always the starting point, and for some, it is also the end point; that is, after reading the abstract very carefully, a busy examiner might only dip into parts of the thesis to clarify various issues/concerns. So, for some examiners, an initial impression based primarily on the abstract will be a primary determinant of an overall opinion of the thesis.

A thesis abstract is also important because it is often used by your university library for indexing purposes, so make sure to include important keywords. The abstract may also be used for disseminating and publicising your work in various ways, and is sometimes used as the basis for presenting an overview of your work for special occasions, such as your graduation ceremony.

Conclusion

A PhD thesis may be the most important document you will ever produce. It is the written evidence of your scientific work, and will be archived permanently in the thesis collection of your university, to be viewed by your university peers, future students, and academics external to your institution. There is no single approach to writing a PhD thesis; presented here is a set of tips derived from my own personal experiences. Others may well advocate alternative approaches.

As I stated at the outset, the process of thesis writing can be arduous or enjoyable. I hope that my set of tips will guide you to write a brilliant PhD thesis – and will help you adopt an approach that will make the process enjoyable.

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